

Immunology

“I hate to disappoint you, but my rubber lips are immune to your charms.”

—Batman & Robin

“No State shall make or enforce any law which shall abridge the privileges or immunities of citizens of the United States . . .”

—The United States Constitution

Mastery of the basic principles and facts in the immunology section will be useful for the Step 1 exam. Cell surface markers are important to know because they are clinically useful (eg, in identifying specific types of immunodeficiency or cancer) and are functionally critical to the jobs immune cells carry out. By spending a little extra effort here, it is possible to turn a traditionally difficult subject into one that is high yield.

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► IMMUNOLOGY—LYMPHOID STRUCTURES

Immune system organs

1° organs:

- Bone marrow—immune cell production, B cell maturation
- Thymus—T cell maturation

2° organs:

- Spleen, lymph nodes, tonsils, Peyer patches
- Allow immune cells to interact with antigen

Lymph node

A 2° lymphoid organ that has many afferents, 1 or more efferents. Encapsulated, with trabeculae. Functions are nonspecific filtration by macrophages, storage of B and T cells, and immune response activation.

Follicle

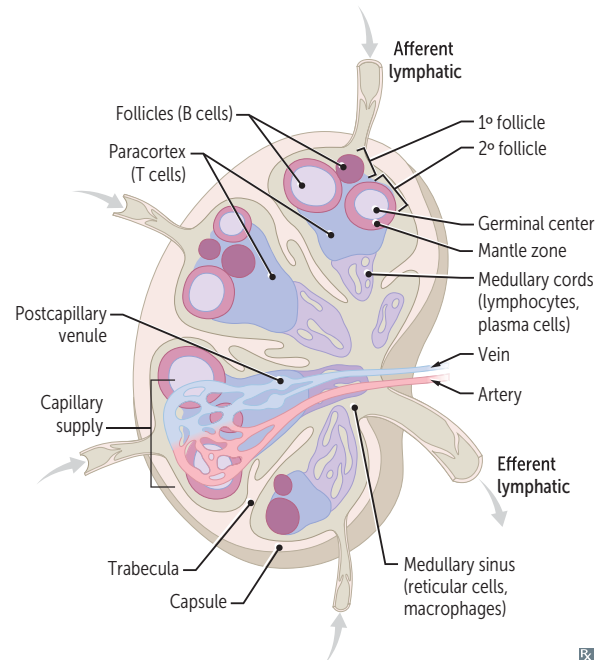
Site of B-cell localization and proliferation. In outer cortex. 1° follicles are dense and dormant. 2° follicles have pale central germinal centers and are active.

Medulla

Consists of medullary cords (closely packed lymphocytes and plasma cells) and medullary sinuses. Medullary sinuses communicate with efferent lymphatics and contain reticular cells and macrophages.

Paracortex

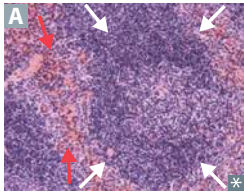
Houses T cells. Region of cortex between follicles and medulla. Contains high endothelial venules through which T and B cells enter from blood. Not well developed in patients with DiGeorge syndrome. Paracortex enlarges in an extreme cellular immune response (eg, viral infection).



Lymph drainage

LYMPH NODE CLUSTER	AREA OF BODY DRAINED
Cervical	Head and neck
Hilar	Lungs
Mediastinal	Trachea and esophagus
Axillary	Upper limb, breast, skin above umbilicus
Celiac	Liver, stomach, spleen, pancreas, upper duodenum
Superior mesenteric	Lower duodenum, jejunum, ileum, colon to splenic flexure
Inferior mesenteric	Colon from splenic flexure to upper rectum
Internal iliac	Lower rectum to anal canal (above pectinate line), bladder, vagina (middle third), cervix, prostate
Para-aortic	Testes, ovaries, kidneys, uterus
Superficial inguinal	Anal canal (below pectinate line), skin below umbilicus (except popliteal area), scrotum, vulva
Popliteal	Dorsolateral foot, posterior calf
Right lymphatic duct drains right side of body above diaphragm.	
Thoracic duct drains everything else into junction of left subclavian and internal jugular veins.	

Spleen



Located in LUQ of abdomen, anterior to left kidney, protected by 9th-11th ribs. Sinusoids are long, vascular channels in red pulp (red arrows in **A**) with fenestrated “barrel hoop” basement membrane.

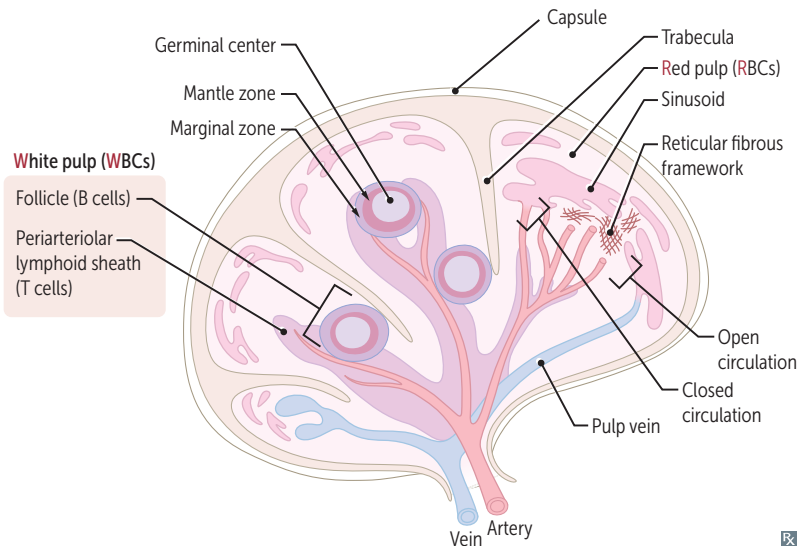
- T cells are found in the periarteriolar lymphatic sheath (PALS) within the white pulp (white arrows in **A**).
- B cells are found in follicles within the white pulp.
- The marginal zone, in between the red pulp and white pulp, contains macrophages and specialized B cells, and is where antigen-presenting cells (APCs) capture blood-borne antigens for recognition by lymphocytes. Macrophages found nearby in spleen remove encapsulated bacteria.

Splenic dysfunction (eg, postsplenectomy, sickle cell disease): ↓ IgM → ↓ complement activation → ↓ C3b opsonization → ↑ susceptibility to encapsulated organisms.

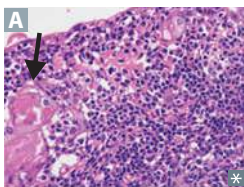
Postsplenectomy:

- Howell-Jolly bodies (nuclear remnants)
- Target cells
- Thrombocytosis (loss of sequestration and removal)
- Lymphocytosis (loss of sequestration)

Vaccinate patients undergoing splenectomy against encapsulated organisms (pneumococcal, Hib, meningococcal).



Thymus



Located in the anterosuperior mediastinum. Site of T-cell differentiation and maturation. Encapsulated. **T**hymus is derived from the **T**hird pharyngeal pouch. Lymphocytes of mesenchymal origin. Cortex is dense with immature T cells; medulla is pale with mature T cells and Hassall corpuscles **A** containing epithelial reticular cells.

T cells = **T**hymus

B cells = **B**one marrow

Hypoplastic in DiGeorge syndrome and severe combined immunodeficiency (SCID).

Thymoma—benign neoplasm of thymus.

Associated with myasthenia gravis and superior vena cava syndrome.

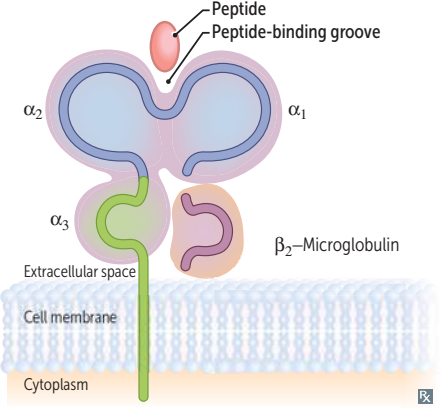
► IMMUNOLOGY—LYMPHOCYTES

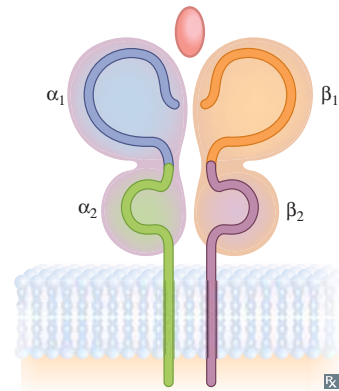
Innate vs adaptive immunity

	Innate immunity	Adaptive immunity
COMPONENTS	Neutrophils, macrophages, monocytes, dendritic cells, natural killer (NK) cells (lymphoid origin), complement	T cells, B cells, circulating antibodies
MECHANISM	Germline encoded	Variation through V(D)J recombination during lymphocyte development
RESISTANCE	Resistance persists through generations; does not change within an organism's lifetime	Microbial resistance not heritable
RESPONSE TO PATHOGENS	Nonspecific Occurs rapidly (minutes to hours) No memory response	Highly specific, refined over time Develops over long periods; memory response is faster and more robust
PHYSICAL BARRIERS	Epithelial tight junctions, mucus	—
SECRETED PROTEINS	Lysozyme, complement, C-reactive protein (CRP), defensins	Immunoglobulins
KEY FEATURES IN PATHOGEN RECOGNITION	Toll-like receptors (TLRs): pattern recognition receptors that recognize pathogen-associated molecular patterns (PAMPs). Examples of PAMPs include LPS (gram $\ominus$ bacteria), flagellin (bacteria), nucleic acids (viruses).	Memory cells: activated B and T cells; subsequent exposure to a previously encountered antigen → stronger, quicker immune response

Major histocompatibility complex I and II

MHC encoded by HLA genes. Present antigen fragments to T cells and bind T-cell receptors (TCRs).

	MHC I	MHC II
LOCI	HLA- A , HLA- B , HLA- C MHC I loci have 1 letter	HLA- DP , HLA- DQ , HLA- DR MHC II loci have 2 letters
BINDING	TCR and CD8	TCR and CD4
STRUCTURE	1 long chain, 1 short chain	2 equal-length chains
EXPRESSION	All nucleated cells, APCs, platelets Not on RBCs	APCs
FUNCTION	Present endogenously synthesized antigens (eg, viral or cytosolic proteins) to CD8+ cytotoxic T cells	Present exogenously synthesized antigens (eg, bacterial proteins) to CD4+ helper T cells
ANTIGEN LOADING	Antigen peptides loaded onto MHC I in RER after delivery via TAP (transporter associated with antigen processing)	Antigen loaded following release of invariant chain in an acidified endosome
ASSOCIATED PROTEINS	β_2 -microglobulin	Invariant chain
STRUCTURE		

**HLA subtypes associated with diseases**

A3	Hemochromatosis	
B8	Addison disease, myasthenia gravis, Graves disease	
B27	P soriatic arthritis, A nkylating spondylitis, I BD-associated arthritis, R eactive arthritis	PAIR . Also known as seronegative arthropathies.
DQ2/DQ8	Celiac disease	I ate (8) too (2) much gluten at Dairy Q ueen.
DR2	M ultiple sclerosis, h ay fever, SLE, Good p asture syndrome	M ultiple hay pastures have d irt.
DR3	Diabetes mellitus type 1, S LE, Graves disease, Hashimoto thyroiditis, Addison disease	2-3, S-L-E
DR4	R heumatoid arthritis, diabetes mellitus type 1, Addison disease	There are 4 walls in a “ r heum” (room).
DR5	Pernicious anemia → vitamin B ₁₂ deficiency, Hashimoto thyroiditis	

Natural killer cells

Lymphocyte member of innate immune system.

Use perforin and granzymes to induce apoptosis of virally infected cells and tumor cells.

Activity enhanced by IL-2, IL-12, IFN- α , and IFN- β .

Induced to kill when exposed to a nonspecific activation signal on target cell and/or to an absence of MHC I on target cell surface.

Also kills via antibody-dependent cell-mediated cytotoxicity (CD16 binds Fc region of bound Ig, activating the NK cell).

Major functions of B and T cells**B cells**

Humoral immunity.

Recognize antigen—undergo somatic hypermutation to optimize antigen specificity.

Produce antibody—differentiate into plasma cells to secrete specific immunoglobulins.

Maintain immunologic memory—memory B cells persist and accelerate future response to antigen.

T cells

Cell-mediated immunity.

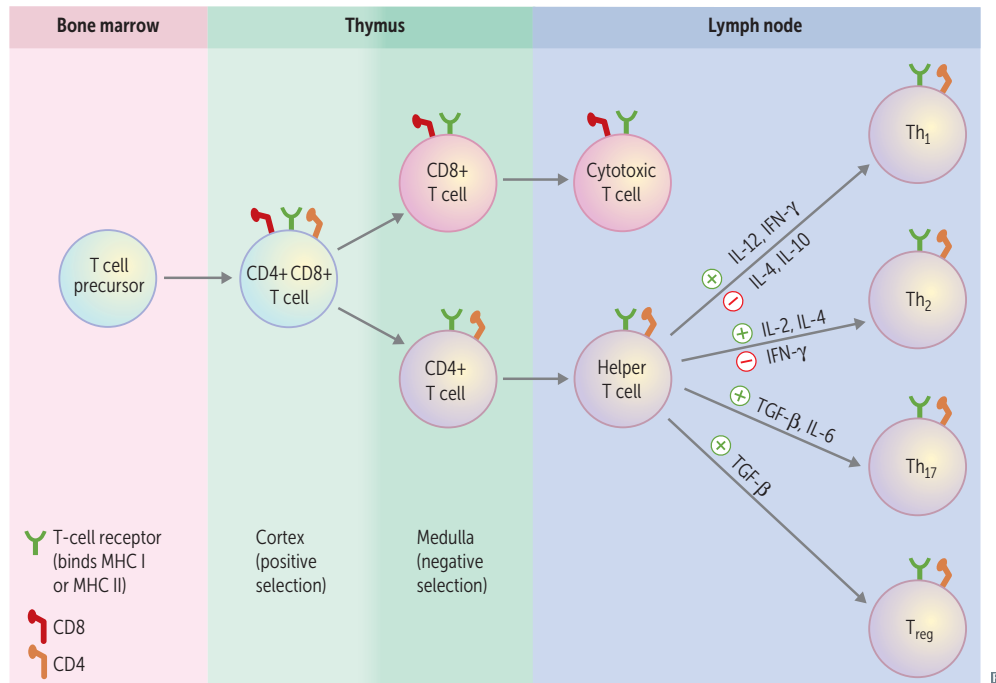
CD4⁺ T cells help B cells make antibodies and produce cytokines to recruit phagocytes and activate other leukocytes.

CD8⁺ T cells directly kill virus-infected cells.

Delayed cell-mediated hypersensitivity (type IV).

Acute and chronic cellular organ rejection.

Rule of 8: MHC II $\times$ CD4 = 8; MHC I $\times$ CD8 = 8.

Differentiation of T cells**Positive selection**

Thymic cortex. T cells expressing TCRs capable of binding self-MHC on cortical epithelial cells survive.

Negative selection

Thymic medulla. T cells expressing TCRs with high affinity for self antigens undergo apoptosis. Tissue-restricted self-antigens are expressed in the thymus due to the action of autoimmune regulator (AIRE); deficiency leads to autoimmune polyendocrine syndrome-1.

Helper T cells**Th1 cell**

Secretes IFN- γ and IL-2

Activates macrophages and cytotoxic T cells

Differentiation induced by IFN- γ and IL-12

Inhibited by IL-4 and IL-10 (from Th2 cell)

Th2 cell

Secretes IL-4, IL-5, IL-6, IL-10, IL-13

Recruits eosinophils for parasite defense and promotes IgE production by B cells

Differentiation induced by IL-2 and IL-4

Inhibited by IFN- γ (from Th1 cell)

Macrophage-lymphocyte interaction—dendritic cells, macrophages, and other APCs release IL-12, which stimulates T cells to differentiate into Th1 cells. Th1 cells release IFN- γ to stimulate macrophages.

Helper T cells have CD4, which binds to MHC II on APCs.

Cytotoxic T cells

Kill virus-infected, neoplastic, and donor graft cells by inducing apoptosis.

Release cytotoxic granules containing preformed proteins (eg, perforin, granzyme B).

Cytotoxic T cells have CD8, which binds to MHC I on virus-infected cells.

Regulatory T cells

Help maintain specific immune tolerance by suppressing CD4 and CD8 T-cell effector functions.

Identified by expression of CD3, CD4, CD25, and FOXP3.

Activated regulatory T cells (Tregs) produce anti-inflammatory cytokines (eg, IL-10, TGF- β).

IPEX (Immune dysregulation, Polyendocrinopathy, Enteropathy, X-linked) syndrome—

genetic deficiency of FOXP3 $\rightarrow$ autoimmunity. Characterized by enteropathy, endocrinopathy, nail dystrophy, dermatitis, and/or other autoimmune dermatologic conditions. Associated with diabetes in male infants.

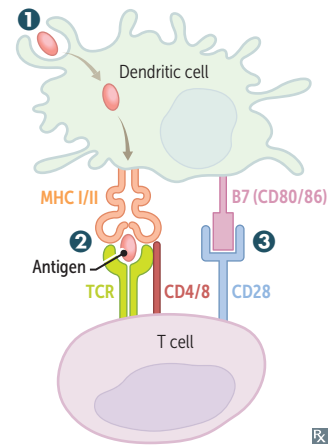
T- and B-cell activation

APCs: B cells, dendritic cells, Langerhans cells, macrophages.

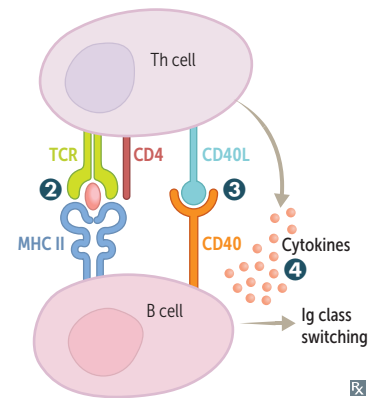
Two signals are required for T-cell activation, B-cell activation, and class switching.

T-cell activation

- ❶ Dendritic cell (specialized APC) samples antigen, processes antigen, and migrates to the draining lymph node.
- ❷ T-cell activation (signal 1): antigen is presented on MHC II and recognized by TCR on Th (CD4+) cell. Endogenous or cross-presented antigen is presented on MHC I to Tc (CD8+) cell.
- ❸ Proliferation and survival (signal 2): costimulatory signal via interaction of B7 protein on dendritic cell (CD80/86) and CD28 on naïve T cell.
- ❹ Th cell activates and produces cytokines. Tc cell activates and is able to recognize and kill virus-infected cell.

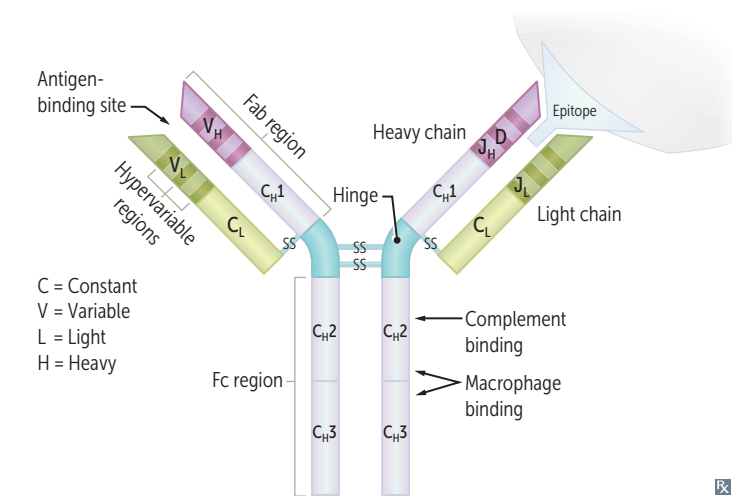
**B-cell activation and class switching**

- ❶ Th-cell activation as above.
- ❷ B-cell receptor-mediated endocytosis; foreign antigen is presented on MHC II and recognized by TCR on Th cell.
- ❸ CD40 receptor on B cell binds CD40 ligand (CD40L) on Th cell.
- ❹ Th cell secretes cytokines that determine Ig class switching of B cell. B cell activates and undergoes class switching, affinity maturation, and antibody production.



Antibody structure and function

Fab (containing the variable/hypervariable regions) consisting of light (L) and heavy (H) chains recognizes antigens. Fc region of IgM and IgG fixes complement. Heavy chain contributes to Fc and Fab regions. Light chain contributes only to Fab region.



Fab:

- **F**ragment, **a**ntigen **b**inding
- Determines idiotype: unique antigen-binding pocket; only 1 antigenic specificity expressed per B cell

Fc:

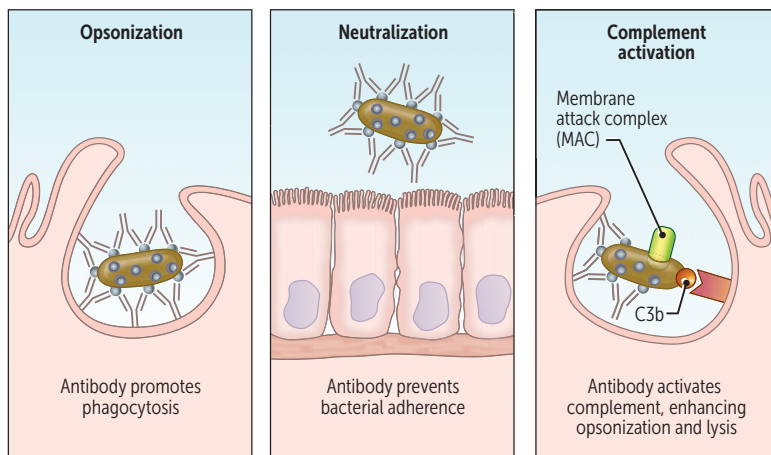
- **C**onstant
- **C**arboxy terminal
- **C**omplement binding
- **C**arbohydrate side chains
- Determines isotype (IgM, IgD, etc)

Generation of antibody diversity (antigen independent)

1. Random recombination of VJ (light-chain) or V(D)J (heavy-chain) genes
2. Random addition of nucleotides to DNA during recombination by terminal deoxynucleotidyl transferase (TdT)
3. Random combination of heavy chains with light chains

Generation of antibody specificity (antigen dependent)

4. Somatic hypermutation and affinity maturation (variable region)
5. Isotype switching (constant region)



Immunoglobulin isotypes

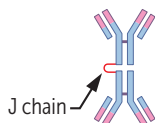
All isotypes can exist as monomers. Mature, naive B cells prior to activation express IgM and IgD on their surfaces. They may differentiate in germinal centers of lymph nodes by isotype switching (gene rearrangement; mediated by cytokines and CD40L) into plasma cells that secrete IgA, IgE, or IgG.

IgG



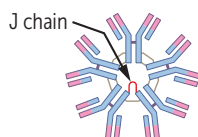
Main antibody in 2° (**delayed**) response to an antigen. Most abundant isotype in serum. Fixes complement, crosses the placenta (provides infants with passive immunity), opsonizes bacteria, neutralizes bacterial toxins and viruses.

IgA



Prevents attachment of bacteria and viruses to mucous membranes; does not fix complement. Monomer (in circulation) or dimer (with J chain when secreted). Crosses epithelial cells by transcytosis. Produced in GI tract (eg, by Peyer patches) and protects against gut infections (eg, *Giardia*). Most produced antibody overall, but has lower serum concentrations. Released into secretions (tears, saliva, mucus) and breast milk. Picks up secretory component from epithelial cells, which protects the Fc portion from luminal proteases.

IgM



Produced in the 1° (**immediate**) response to an antigen. Fixes complement but does not cross the placenta. Antigen receptor on the surface of B cells. Monomer on B cell, pentamer with J chain when secreted. Pentamer enables avid binding to antigen while humoral response evolves.

IgD



Unclear function. Found on surface of many B cells and in serum.

IgE



Binds mast cells and basophils; cross-links when exposed to allergen, mediating immediate (type I) hypersensitivity through release of inflammatory mediators such as histamine. Contributes to immunity to worms by activating eosinophils. Lowest concentration in serum.

Antigen type and memory

Thymus-independent antigens

Antigens lacking a peptide component (eg, lipopolysaccharides from gram $\ominus$ bacteria); cannot be presented by MHC to T cells. Weakly immunogenic; vaccines often require boosters and adjuvants (eg, pneumococcal polysaccharide vaccine).

Thymus-dependent antigens

Antigens containing a protein component (eg, diphtheria vaccine). Class switching and immunologic memory occur as a result of direct contact of B cells with Th cells.

▶ IMMUNOLOGY—IMMUNE RESPONSES

Acute-phase reactants Factors whose serum concentrations change significantly in response to inflammation; produced by the liver in both acute and chronic inflammatory states. Notably induced by IL-6.

POSITIVE (UPREGULATED)

C-reactive protein

Opsonin; fixes complement and facilitates phagocytosis.
Measured clinically as a nonspecific sign of ongoing inflammation.

Ferritin

Binds and sequesters iron to inhibit microbial iron scavenging.

Fibrinogen

Coagulation factor; promotes endothelial repair; correlates with ESR.

Hepcidin

↓ iron absorption (by degrading ferroportin) and ↓ iron release (from macrophages) → anemia of chronic disease.

Serum amyloid A

Prolonged elevation can lead to amyloidosis.

NEGATIVE (DOWNREGULATED)

Albumin

Reduction conserves amino acids for positive reactants.

Transferrin

Internalized by macrophages to sequester iron.

Complement

System of hepatically synthesized plasma proteins that play a role in innate immunity and inflammation. Membrane attack complex (MAC) defends against gram $\ominus$ bacteria.

ACTIVATION

Classic pathway—IgG or IgM mediated.

GM makes **classic** cars.

Alternative pathway—microbe surface molecules.

Lectin pathway—mannose or other sugars on microbe surface.

FUNCTIONS

C3b—opsonization.

C3b **b**inds **b**acteria.

C3a, C4a, C5a—**a**naphylaxis.

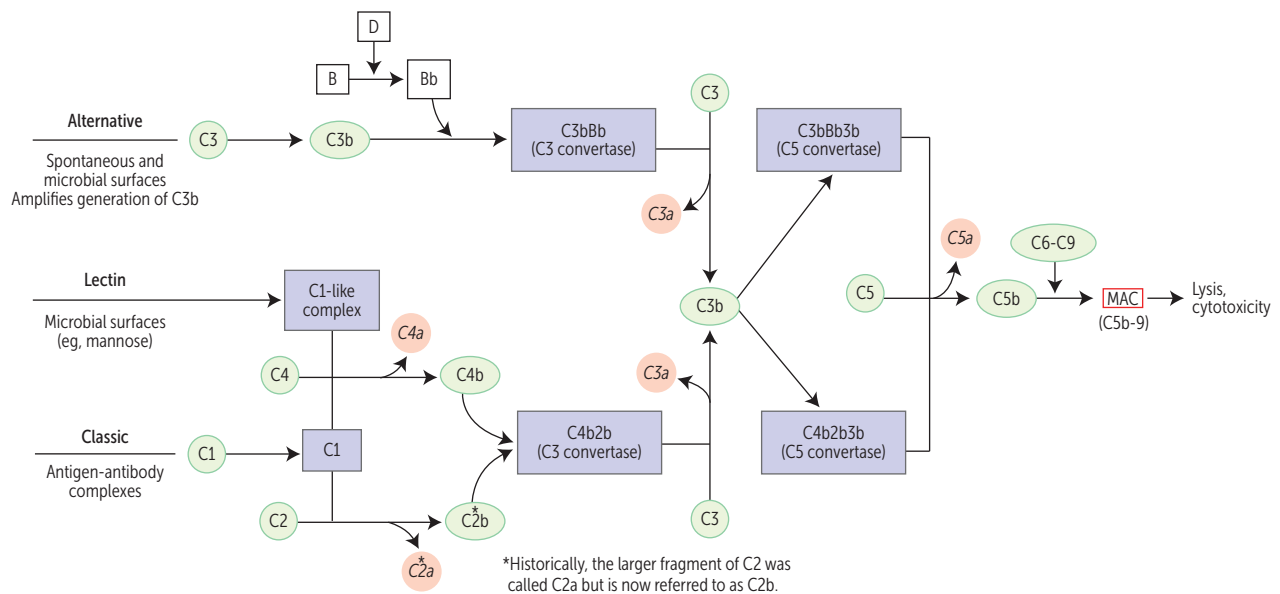
C5a—neutrophil chemotaxis.

C5b-9—cytolysis by MAC.

Opsonins—C3b and IgG are the two 1^o opsonins in bacterial defense; enhance phagocytosis. C3b also helps clear immune complexes.

Opsonin (Greek) = to prepare for eating.

Inhibitors—decay-accelerating factor (DAF, aka CD55) and C1 esterase inhibitor help prevent complement activation on self cells (eg, RBCs).

**Complement disorders****Complement protein deficiencies****C3 deficiency**

Increases risk of severe, recurrent pyogenic sinus and respiratory tract infections; ↑ susceptibility to type III hypersensitivity reactions.

C5–C9 deficiencies

Terminal complement deficiency increases susceptibility to recurrent *Neisseria* bacteremia.

Complement regulatory protein deficiencies**C1 esterase inhibitor deficiency**

Causes hereditary angioedema due to unregulated activation of kallikrein → ↑ bradykinin. Characterized by ↓ C4 levels. ACE inhibitors are contraindicated.

CD55 deficiency

Also called decay-accelerating factor (DAF) deficiency. Causes complement-mediated lysis of RBCs and paroxysmal nocturnal hemoglobinuria.

Important cytokines

SECRETED BY MACROPHAGES

Interleukin-1

Causes fever, acute inflammation. Activates endothelium to express adhesion molecules. Induces chemokine secretion to recruit WBCs.

“**Hot T-bone stEAK**”:

IL-1: fever (**hot**).
 IL-2: stimulates **T** cells.
 IL-3: stimulates **bone** marrow.
 IL-4: stimulates Ig**E** production.
 IL-5: stimulates Ig**A** production.
 IL-6: stimulates a**K**ute-phase protein production.

Interleukin-6

Causes fever and stimulates production of acute-phase proteins.

Interleukin-8

Major chemotactic factor for neutrophils.

“**Clean up on aisle 8.**” Neutrophils are recruited by **IL-8** to **clear** infections.

Interleukin-12

Induces differentiation of T cells into Th1 cells. Activates NK cells.

Tumor necrosis factor- α

Activates endothelium. Causes WBC recruitment, vascular leak.

Causes cachexia in malignancy. Maintains granulomas in TB. IL-1, IL-6, and TNF- α can mediate sepsis.

SECRETED BY ALL T CELLS

Interleukin-2

Stimulates growth of helper, cytotoxic, and regulatory T cells, and NK cells.

Interleukin-3

Supports growth and differentiation of bone marrow stem cells. Functions like GM-CSF.

FROM Th1 CELLS

Interferon- γ

Secreted by NK cells and T cells in response to antigen or IL-12 from macrophages; stimulates macrophages to kill phagocytosed pathogens. Inhibits differentiation of Th2 cells.

Also activates NK cells to kill virus-infected cells. Increases MHC expression and antigen presentation by all cells.

FROM Th2 CELLS

Interleukin-4

Induces differentiation of T cells into Th2 cells. Promotes growth of B cells. Enhances class switching to IgE and IgG.

Interleukin-5

Promotes growth and differentiation of B cells. Enhances class switching to IgA. Stimulates growth and differentiation of eosinophils.

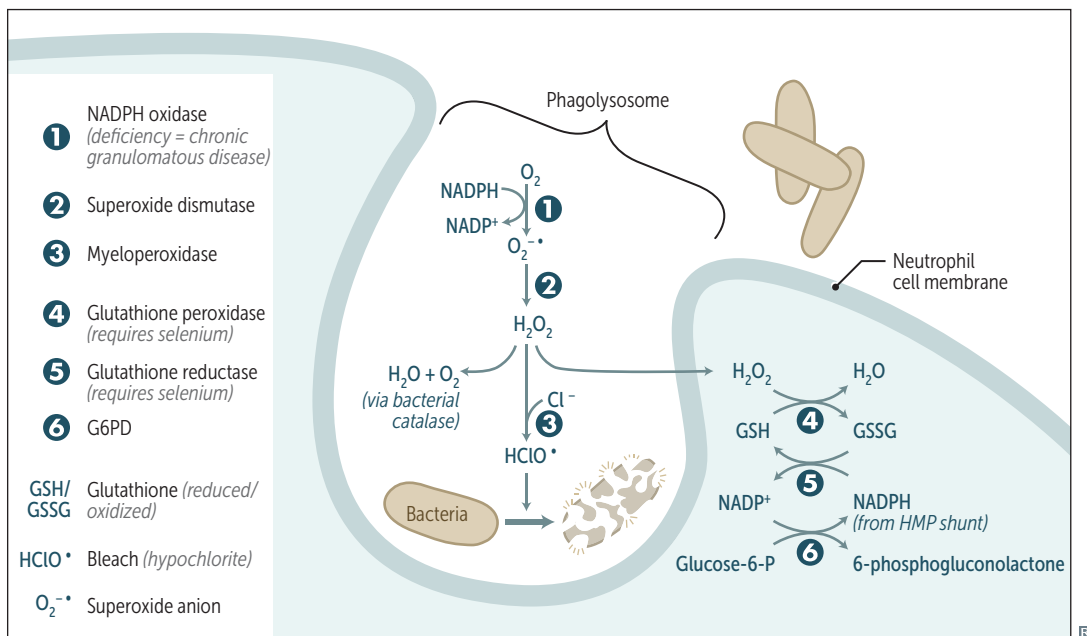
Interleukin-10

Attenuates inflammatory response. Decreases expression of MHC class II and Th1 cytokines. Inhibits activated macrophages and dendritic cells. Also secreted by regulatory T cells.

TGF- β and IL-10 both **attenuate** the immune response.

Respiratory burst (oxidative burst)

Involves the activation of the phagocyte NADPH oxidase complex (eg, in neutrophils, monocytes), which utilizes O_2 as a substrate. Plays an important role in the immune response → rapid release of reactive oxygen species (ROS). NADPH plays a role in both the creation and neutralization of ROS. Myeloperoxidase is a blue-green heme-containing pigment that gives sputum its color.



Phagocytes of patients with CGD can utilize H_2O_2 generated by invading organisms and convert it to ROS. Patients are at ↑ risk for infection by catalase ⊕ species (eg, *S aureus*, *Aspergillus*) capable of neutralizing their own H_2O_2 , leaving phagocytes without ROS for fighting infections.

Pyocyanin of *P aeruginosa* functions to generate ROS to kill competing microbes. Lactoferrin is a protein found in secretory fluids and neutrophils that inhibits microbial growth via iron chelation.

Interferon-α and -β

A part of innate host defense against both RNA and DNA viruses. **Interferons** are glycoproteins synthesized by virus-infected cells that act locally on uninfected cells, “priming them” for viral defense by helping to degrade viral nucleic acid and protein.

Interfere with viruses.

Cell surface proteins	MHC I present on all nucleated cells (ie, not mature RBCs).	
T cells	TCR (binds antigen-MHC complex) CD3 (associated with TCR for signal transduction) CD28 (binds B7 on APC) CXCR4/CCR5 (co-receptors for HIV)	
Helper T cells	CD4, CD40L	
Cytotoxic T cells	CD8 CXCR4/CCR5	
Regulatory T cells	CD4, CD25	
B cells	Ig (binds antigen) CD19, CD20, CD21 (receptor for EBV), CD40 MHC II, B7	You can drink B eer at the B ar when you're 21 : B cells, Epstein- B arr virus, CD21 .
Macrophages	CD14 (receptor for PAMPs, eg, LPS), CD40 CCR5 MHC II, B7 (CD80/86) Fc and C3b receptors (enhanced phagocytosis)	
NK cells	CD56 (suggestive marker for NK)	
Hematopoietic stem cells	CD34	

Anergy State during which a cell cannot become activated by exposure to its antigen. T and B cells become anergic when exposed to their antigen without costimulatory signal (signal 2). Another mechanism of self-tolerance.

Effects of bacterial toxins Superantigens (*S pyogenes* and *S aureus*)—cross-link the β region of the T-cell receptor to the MHC class II on APCs. Can activate any CD4+ T cell → massive release of cytokines.
Endotoxins/lipopolysaccharide (gram $\ominus$ bacteria)—directly stimulate macrophages by binding to endotoxin receptor TLR4/CD14; Th cells are not involved.

Antigenic variation Classic examples:

- Bacteria—*Salmonella* (2 flagellar variants), *Borrelia recurrentis* (relapsing fever), *N gonorrhoeae* (pilus protein)
- Viruses—influenza, HIV, HCV
- Parasites—trypanosomes

Some mechanisms for variation include DNA rearrangement and RNA segment reassortment (eg, influenza major shift) or protein mutations (eg, influenza minor drift).

Passive vs active immunity

	Passive	Active
MEANS OF ACQUISITION	Receiving preformed antibodies	Exposure to foreign antigens
ONSET	Rapid	Slow
DURATION	Short span of antibodies (half-life = 3 weeks)	Long-lasting protection (memory)
EXAMPLES	IgA in breast milk, maternal IgG crossing placenta, antitoxin, humanized monoclonal antibody	Natural infection, vaccines, toxoid
NOTES	After exposure to T etanus toxin, B otulinum toxin, H BV, V aricella, R abies virus, or diphtheria antitoxin, unvaccinated patients are given preformed antibodies (passive)—“ T o B e H ealed V ery R apidly”	Combined passive and active immunizations can be given for hepatitis B or rabies exposure

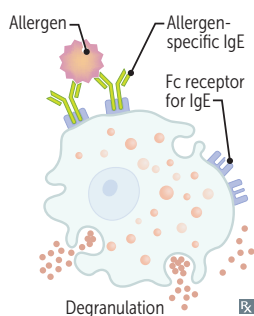
Vaccination

Induces an active immune response (humoral and/or cellular) to specific pathogens.

VACCINE TYPE	DESCRIPTION	PROS/CONS	EXAMPLES
Live attenuated vaccine	Microorganism loses its pathogenicity but retains capacity for transient growth within inoculated host. Induces cellular and humoral responses . MMR and varicella are live vaccines that can be given to patients with HIV who have a CD4 cell count > 200/mm ³ .	Pro: induces strong, often lifelong immunity. Con: may revert to virulent form. Often contraindicated in pregnancy and immunodeficiency.	BCG, influenza (intranasal), measles, mumps, polio (Sabin), rotavirus, rubella, varicella, yellow fever.
Inactivated or killed vaccine	Pathogen is inactivated by heat or chemicals. Maintaining epitope structure on surface antigens is important for immune response. Mainly induces a humoral response .	Pro: safer than live vaccines. Con: weaker immune response; booster shots usually required.	R abies, I nfluenza (injection), P olio (Salk), hepatitis A (“ R.I.P. Always ”).

Hypersensitivity types Four types: **A**naphylactic and **A**topic (type I), **C**ytotoxic (antibody mediated, type II), **I**mmune complex (type III), **D**elayed (cell mediated, type IV) (**ACID**).

Type I



Anaphylactic and atopic—free antigen cross-links IgE on presensitized mast cells and basophils, triggering immediate release of vasoactive amines that act at postcapillary venules (ie, histamine). Reaction develops rapidly after antigen exposure because of preformed antibody. Delayed phase results from mast cells and basophils releasing cytokines that induce cellular inflammation.

First (type) and **F**ast (anaphylaxis).

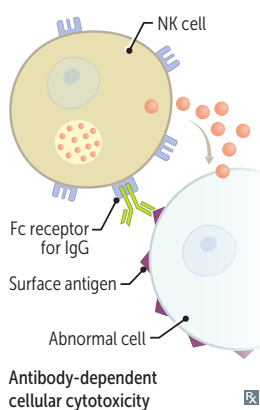
Types I, II, and III are all antibody mediated.

Test: skin test or blood test (ELISA) for allergen-specific IgE.

Example:

- Anaphylaxis (eg, food, drug, or bee sting allergies)

Type II



Antibodies bind to cell-surface antigens → cellular destruction, inflammation, and cellular dysfunction.

Cellular destruction: cell is opsonized (coated) by antibodies, leading to either:

- Phagocytosis and/or activation of complement system.
- NK cell killing (antibody-dependent cellular cytotoxicity).

Inflammation—binding of antibodies to cell surfaces → activation of complement system and Fc receptor-mediated inflammation.

Cellular dysfunction—antibodies bind to cell surface receptors → abnormal blockade or activation of downstream process.

Direct Coombs test—detects antibodies attached **directly** to the RBC surface.

Indirect Coombs test—detects presence of unbound antibodies in the serum

Examples:

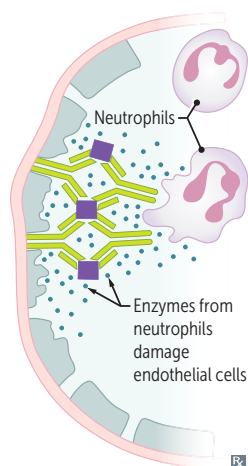
- Autoimmune-hemolytic anemia
- Immune thrombocytopenic purpura
- Transfusion reactions
- Hemolytic disease of the newborn

Examples:

- Goodpasture syndrome
- Rheumatic fever
- Hyperacute transplant rejection

Examples:

- Myasthenia gravis
- Graves disease

Hypersensitivity types (continued)**Type III**

Immune complex—antigen-antibody (IgG) complexes activate complement, which attracts neutrophils; neutrophils release lysosomal enzymes.

Can be associated with vasculitis and systemic manifestations.

Serum sickness—an immune complex disease in which antibodies to foreign proteins are produced (takes 5 days). Immune complexes form and are deposited in membranes, where they fix complement (leads to tissue damage). More common than Arthus reaction.

Arthus reaction—a local subacute antibody-mediated hypersensitivity reaction. Intradermal injection of antigen into a presensitized (has circulating IgG) individual leads to immune complex formation in the skin. Characterized by edema, necrosis, and activation of complement.

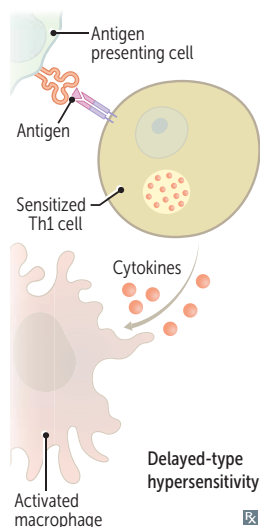
In type **III** reaction, imagine an immune complex as **3** things stuck together: antigen-antibody-complement.

Examples:

- SLE
- Polyarteritis nodosa
- Poststreptococcal glomerulonephritis

Most serum sickness is now caused by drugs (not serum) acting as haptens. Fever, urticaria, arthralgia, proteinuria, lymphadenopathy occur 5–10 days after antigen exposure.

Antigen-antibody complexes cause the Arthus reaction.

Type IV

Two mechanisms, each involving T cells:

1. Direct cell cytotoxicity: CD8+ cytotoxic T cells kill targeted cells.
2. Delayed-type hypersensitivity: sensitized CD4+ helper T cells encounter antigen and release cytokines → inflammation and macrophage activation.

Response does not involve antibodies (vs types I, II, and III).

Example:

- Type 1 diabetes mellitus

Examples:

- Contact dermatitis (eg, poison ivy, nickel allergy)
- Graft-versus-host disease

Tests: PPD, patch test.

4T's: **T** cells, **T**ransplant rejections, **T**B skin tests, **T**ouching (contact dermatitis).

Fourth (type) and **last** (delayed).

Blood transfusion reactions

TYPE	PATHOGENESIS	CLINICAL PRESENTATION	TIMING
Allergic/anaphylactic reaction	Type I hypersensitivity reaction against plasma proteins in transfused blood. IgA-deficient individuals must receive blood products without IgA.	Urticaria, pruritus, fever, wheezing, hypotension, respiratory arrest, shock.	Within minutes to 2–3 hours
Febrile nonhemolytic transfusion reaction	Type II hypersensitivity reaction. Host antibodies against donor HLA antigens and WBCs.	Fever, headaches, chills, flushing.	Within 1–6 hours
Acute hemolytic transfusion reaction	Type II hypersensitivity reaction. Intravascular hemolysis (ABO blood group incompatibility) or extravascular hemolysis (host antibody reaction against foreign antigen on donor RBCs).	Fever, hypotension, tachypnea, tachycardia, flank pain, hemoglobinuria (intravascular hemolysis), jaundice (extravascular).	Within 1 hour
Transfusion-related acute lung injury	Donor anti-leukocyte antibodies against recipient neutrophils and pulmonary endothelial cells.	Respiratory distress and noncardiogenic pulmonary edema.	Within 6 hours

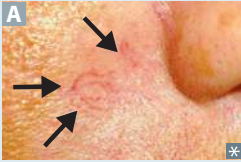
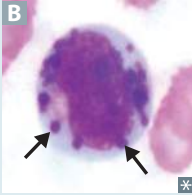
Autoantibodies

AUTOANTIBODY	ASSOCIATED DISORDER
Anti-ACh receptor	Myasthenia gravis
Anti-glomerular basement membrane	Goodpasture syndrome
Anti- β_2 glycoprotein	Antiphospholipid syndrome
Anticardiolipin, lupus anticoagulant	SLE, antiphospholipid syndrome
Anticentromere	Limited scleroderma (CREST syndrome)
Anti-desmoglein (anti-desmosome)	Pemphigus vulgaris
Anti-glutamic acid decarboxylase, islet cell cytoplasmic antibodies	Type 1 diabetes mellitus
Anti-hemidesmosome	Bullous pemphigoid
Antisynthetase (eg, anti-Jo-1), anti-SRP, anti-helicase (anti-Mi-2)	Polymyositis, dermatomyositis
Antimicrosomal, antithyroglobulin, anti-thyroid peroxidase	Hashimoto thyroiditis
Antimitochondrial	1° biliary cirrhosis
Antiparietal cell, anti-intrinsic factor	Pernicious anemia
Antiphospholipase A ₂ receptor	1° membranous nephropathy
Anti- Scl -70 (anti-DNA topoisomerase I)	Scl eroderma (diffuse)
Anti-smooth muscle	Autoimmune hepatitis type 1
Anti- SSA , anti- SSB (anti-Ro, anti-La)	Sj ögren s ndrome
Anti-TSH receptor	Graves disease
Anti-presynaptic voltage-gated calcium channel	Lambert-Eaton myasthenic syndrome
IgA anti-endomysial, IgA anti-tissue transglutaminase	Celiac disease
MPO-ANCA/p-ANCA	Microscopic polyangiitis, eosinophilic granulomatosis with polyangiitis (Churg-Strauss syndrome), ulcerative colitis
PR3-ANCA/c-ANCA	Granulomatosis with polyangiitis (Wegener)
Rheumatoid factor (IgM antibody against IgG Fc region), anti-CCP (more specific)	Rheumatoid arthritis
Antinuclear (ANA)	Nonspecific screening antibody, often associated with SLE
Anti-dsDNA, anti-Smith	SLE
Anti-histone	Drug-induced lupus
Anti-U1 RNP (ribonucleoprotein)	Mixed connective tissue disease

Immunodeficiencies

DISEASE	DEFECT	PRESENTATION	FINDINGS
B-cell disorders			
X-linked (Bruton) agammaglobulinemia	Defect in BTK , a tyrosine kinase gene → no B -cell maturation. X-linked recessive (↑ in B oys).	Recurrent bacterial and enteroviral infections after 6 months (↓ maternal IgG).	Absent B cells in peripheral blood, ↓ Ig of all classes. Absent/scanty lymph nodes and tonsils. Live vaccines contraindicated.
Selective IgA deficiency	Unknown. Most common 1° immunodeficiency.	Majority A symptomatic. Can see A irway and GI infections, A utoimmune disease, A topy, A naphylaxis to Ig A -containing products.	↓ IgA with normal IgG, IgM levels. ↑ susceptibility to giardiasis.
Common variable immunodeficiency	Defect in B-cell differentiation. Many causes.	Usually presents after age 2 and may be considerably delayed; ↑ risk of autoimmune disease, bronchiectasis, lymphoma, sinopulmonary infections.	↓ plasma cells, ↓ immunoglobulins.
T-cell disorders			
Thymic aplasia (DiGeorge syndrome)	22q11 deletion; failure to develop 3rd and 4th pharyngeal pouches → absent thymus and parathyroids.	Tetany (hypocalcemia), recurrent viral/fungal infections (T-cell deficiency), conotruncal abnormalities (eg, tetralogy of Fallot, truncus arteriosus).	↓ T cells, ↓ PTH, ↓ Ca ²⁺ . Absent thymic shadow on CXR.
IL-12 receptor deficiency	↓ Th1 response. Autosomal recessive.	Disseminated mycobacterial and fungal infections; may present after administration of BCG vaccine.	↓ IFN-γ.
Autosomal dominant hyper-IgE syndrome (Job syndrome)	Deficiency of Th17 cells due to STAT3 mutation → impaired recruitment of neutrophils to sites of infection.	FATED : coarse F acies, cold (noninflamed) staphylococcal A bscesses, retained primary T eeth, ↑ Ig E , D ermatologic problems (eczema). Bone fractures from minor trauma.	↑ IgE, ↓ IFN-γ. ↑ eosinophils.
Chronic mucocutaneous candidiasis	T-cell dysfunction. Many causes.	Noninvasive <i>Candida albicans</i> infections of skin and mucous membranes.	Absent in vitro T-cell proliferation in response to <i>Candida</i> antigens. Absent cutaneous reaction to <i>Candida</i> antigens.

Immunodeficiencies (continued)

DISEASE	DEFECT	PRESENTATION	FINDINGS
B- and T-cell disorders			
Severe combined immunodeficiency	Several types including defective IL-2R gamma chain (most common, X-linked), adenosine deaminase deficiency (autosomal recessive).	Failure to thrive, chronic diarrhea, thrush. Recurrent viral, bacterial, fungal, and protozoal infections. Treatment: avoid live vaccines, give antimicrobial prophylaxis and IVIG; bone marrow transplant curative (no concern for rejection).	↓ T-cell receptor excision circles (TRECs). Absence of thymic shadow (CXR), germinal centers (lymph node biopsy), and T cells (flow cytometry).
Ataxia-telangiectasia 	Defects in ATM gene → failure to repair DNA double strand breaks → cell cycle arrest.	Triad: cerebellar defects (A taxia), spider A ngiomas (telangiectasia A), Ig A deficiency.	↑ A FP. ↓ IgA, IgG, and IgE. Lymphopenia, cerebellar atrophy. ↑ risk of lymphoma and leukemia.
Hyper-IgM syndrome	Most commonly due to defective CD40L on Th cells → class switching defect; X-linked recessive.	Severe pyogenic infections early in life; opportunistic infection with <i>Pneumocystis</i> , <i>Cryptosporidium</i> , CMV.	Normal or ↑ IgM. ↓↓ IgG, IgA, IgE. Failure to make germinal centers.
Wiskott-Aldrich syndrome	Mutation in WASp gene; leukocytes and platelets unable to reorganize actin cytoskeleton → defective antigen presentation. X-linked recessive.	WATER: Wiskott-Aldrich: T hrombocytopenia, E czema, R ecurrent (pyogenic) infections. ↑ risk of autoimmune disease and malignancy.	↓ to normal IgG, IgM. ↑ IgE, IgA. Fewer and smaller platelets.
Phagocyte dysfunction			
Leukocyte adhesion deficiency (type 1)	Defect in LFA-1 integrin (CD18) protein on phagocytes; impaired migration and chemotaxis; autosomal recessive.	Recurrent skin and mucosal bacterial infections, absent pus, impaired wound healing, delayed (> 30 days) separation of umbilical cord.	↑ neutrophils. Absence of neutrophils at infection sites.
Chédiak-Higashi syndrome 	Defect in lysosomal trafficking regulator gene (LYST). Microtubule dysfunction in phagosome-lysosome fusion; autosomal recessive.	Recurrent pyogenic infections by staphylococci and streptococci, partial albinism, peripheral neuropathy, progressive neurodegeneration, infiltrative lymphohistiocytosis.	Giant granules (B , arrows) in granulocytes and platelets. Pancytopenia. Mild coagulation defects.
Chronic granulomatous disease	Defect of NADPH oxidase → ↓ reactive oxygen species (eg, superoxide) and ↓ respiratory burst in neutrophils; X-linked recessive most common.	↑ susceptibility to catalase ⊕ organisms.	Abnormal dihydrorhodamine (flow cytometry) test (↓ green fluorescence). Nitroblue tetrazolium dye reduction test (obsolete) fails to turn blue.

Infections in immunodeficiency

PATHOGEN	↓ T CELLS	↓ B CELLS	↓ GRANULOCYTES	↓ COMPLEMENT
Bacteria	Sepsis	Encapsulated (Please SHINE my SKiS): P <i>seudomonas aeruginosa</i> , S <i>treptococcus pneumoniae</i> , H <i>aeophilus</i> I <i>nfluenzae</i> type B, N <i>eisseria meningitidis</i> , E <i>scherichia coli</i> , S <i>almonella</i> , K <i>lebsiella pneumoniae</i> , Group B S <i>treptococcus</i>	<i>Staphylococcus</i> , <i>Burkholderia cepacia</i> , <i>Pseudomonas aeruginosa</i> , <i>Serratia</i> , <i>Nocardia</i>	Encapsulated species with early component deficiencies <i>Neisseria</i> with late complement (C5–C9) deficiencies
Viruses	CMV, EBV, JC virus, VZV, chronic infection with respiratory/GI viruses	Enteroviral encephalitis, poliovirus (live vaccine contraindicated)	N/A	N/A
Fungi/parasites	<i>Candida</i> (local), PCP, <i>Cryptococcus</i>	GI giardiasis (no IgA)	<i>Candida</i> (systemic), <i>Aspergillus</i> , <i>Mucor</i>	N/A

Note: **B**-cell deficiencies tend to produce recurrent **b**acterial infections, whereas T-cell deficiencies produce more fungal and viral infections.

Grafts

Autograft	From self.
Syngeneic graft (isograft)	From identical twin or clone.
Allograft	From nonidentical individual of same species.
Xenograft	From different species.

Transplant rejection

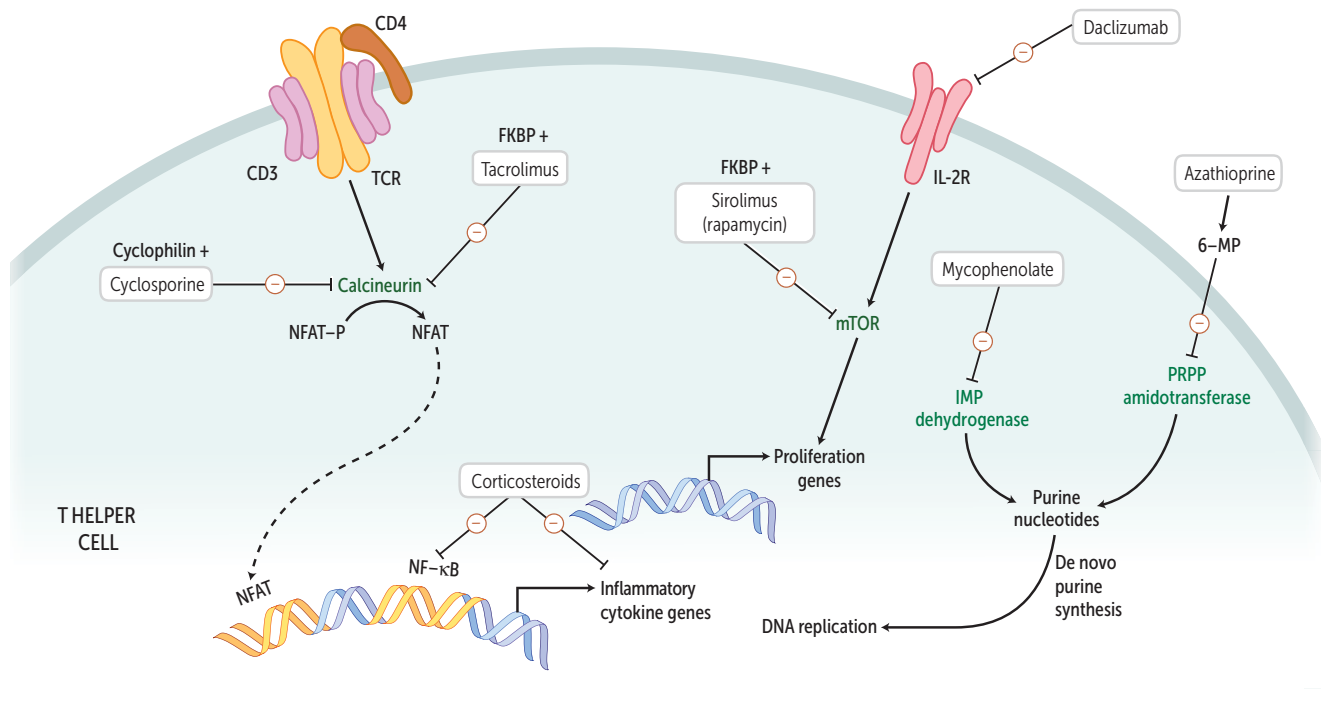
TYPE OF REJECTION	ONSET	PATHOGENESIS	FEATURES
Hyperacute	Within minutes	Pre-existing recipient antibodies react to donor antigen (type II hypersensitivity reaction), activate complement.	Widespread thrombosis of graft vessels → ischemia/necrosis. Graft must be removed.
Acute	Weeks to months	Cellular: CD8+ T cells activated against donor MHCs (type IV hypersensitivity reaction). Humoral: similar to hyperacute, except antibodies develop after transplant.	Vasculitis of graft vessels with dense interstitial lymphocytic infiltrate. Prevent/reverse with immunosuppressants.
Chronic	Months to years	CD4+ T cells respond to recipient APCs presenting donor peptides, including allogeneic MHC. Both cellular and humoral components (type II and IV hypersensitivity reactions).	Recipient T cells react and secrete cytokines → proliferation of vascular smooth muscle, parenchymal atrophy, interstitial fibrosis. Dominated by arteriosclerosis. Organ-specific examples: ▪ Bronchiolitis obliterans (lung) ▪ Accelerated atherosclerosis (heart) ▪ Chronic graft nephropathy (kidney) ▪ Vanishing bile duct syndrome (liver)
Graft-versus-host disease	Varies	Grafted immunocompetent T cells proliferate in the immunocompromised host and reject host cells with “foreign” proteins → severe organ dysfunction. Type IV hypersensitivity reaction.	Maculopapular rash, jaundice, diarrhea, hepatosplenomegaly. Usually in bone marrow and liver transplants (rich in lymphocytes). Potentially beneficial in bone marrow transplant for leukemia (graft-versus-tumor effect).

► IMMUNOLOGY—IMMUNOSUPPRESSANTS

Immunosuppressants Agents that block lymphocyte activation and proliferation. Reduce acute transplant rejection by suppressing cellular immunity (used as prophylaxis). Frequently combined to achieve greater efficacy with ↓ toxicity. Chronic suppression ↑ risk of infection and malignancy.

DRUG	MECHANISM	OTHER USE	TOXICITY	NOTES
Cyclosporine	Calcineurin inhibitor; binds cyclophilin . Blocks T-cell activation by preventing IL-2 transcription .	Psoriasis, rheumatoid arthritis.	Nephrotoxicity , hypertension, hyperlipidemia, neurotoxicity, gingival hyperplasia, hirsutism.	Both calcineurin inhibitors are highly nephrotoxic.
Tacrolimus (FK506)	Calcineurin inhibitor; binds FK506 binding protein (FKBP). Blocks T-cell activation by preventing IL-2 transcription .		Similar to cyclosporine, ↑ risk of diabetes and neurotoxicity; no gingival hyperplasia or hirsutism.	
Sirolimus (Rapamycin)	mTOR inhibitor; binds FKBP. Blocks T-cell activation and B-cell differentiation by preventing response to IL-2 .	Kidney transplant rejection prophylaxis specifically.	“Pan Sir topenia” (pancytopenia), insulin resistance, hyperlipidemia; not nephrotoxic .	Kidney “ sir -vives.” Synergistic with cyclosporine. Also used in drug-eluting stents.
Basiliximab	Monoclonal antibody; blocks IL-2R.		Edema, hypertension, tremor.	
Azathioprine	Antimetabolite precursor of 6-mercaptopurine. Inhibits lymphocyte proliferation by blocking nucleotide synthesis.	Rheumatoid arthritis, Crohn disease, glomerulonephritis, other autoimmune conditions.	Pancytopenia.	6-MP degraded by xanthine oxidase; toxicity ↑ by allopurinol. Pronounce “azathio- purine .”
Mycophenolate mofetil	Reversibly inhibits IMP dehydrogenase, preventing purine synthesis of B and T cells.	Lupus nephritis.	GI upset, pancytopenia, hypertension, hyperglycemia. Less nephrotoxic and neurotoxic.	Associated with invasive CMV infection.
Corticosteroids	Inhibit NF-κB. Suppress both B- and T-cell function by ↓ transcription of many cytokines. Induce T cell apoptosis.	Many autoimmune and inflammatory disorders, adrenal insufficiency, asthma, CLL, non-Hodgkin lymphoma.	Cushing syndrome, osteoporosis, hyperglycemia, diabetes, amenorrhea, adrenocortical atrophy, peptic ulcers, psychosis, cataracts, avascular necrosis (femoral head).	Demargination of WBCs causes artificial leukocytosis. Adrenal insufficiency may develop if drug is stopped abruptly after chronic use.

Immunosuppression targets



Recombinant cytokines and clinical uses

AGENT	CLINICAL USES
Aldesleukin (IL-2)	Renal cell carcinoma, metastatic melanoma
Epoetin alfa (erythropoietin)	Anemias (especially in renal failure)
Filgrastim (G-CSF)	Recovery of bone marrow and WBC counts by granulocyte stimulation
Sargramostim (GM-CSF)	Recovery of bone marrow and WBC counts by granulocyte and monocyte stimulation
IFN-α	Chronic hepatitis B and C, Kaposi sarcoma, malignant melanoma, hairy cell leukemia, condyloma acuminata, renal cell carcinoma
IFN-β	Multiple sclerosis
IFN-γ	Chronic granulomatous disease
Romiplostim (thrombopoietin analog), eltrombopag (thrombopoietin receptor agonist)	Thrombocytopenia
Oprelvekin (IL-11)	Thrombocytopenia

Therapeutic antibodies

AGENT	TARGET	CLINICAL USE	NOTES
Cancer therapy			
Alemtuzumab	CD52	CLL, MS	“ Aly mtuzumab”—chronic ly mphocytic leukemia
Bevacizumab	VEGF	Colorectal cancer, renal cell carcinoma, non-small cell lung cancer	
Cetuximab	EGFR	Stage IV colorectal cancer, head and neck cancer	
Rituximab	CD20	B-cell non-Hodgkin lymphoma, CLL, rheumatoid arthritis, ITP	
Trastuzumab	HER2/neu	Breast cancer, gastric cancer	HER 2 —“tra s2 zumab”
Autoimmune disease therapy			
Adalimumab, certolizumab, golimumab, infliximab	Soluble TNF- α	IBD, rheumatoid arthritis, ankylosing spondylitis, psoriasis	Etanercept is a decoy TNF- α receptor and not a monoclonal antibody
Daclizumab	CD25 (part of IL-2 receptor)	Relapsing multiple sclerosis	
Eculizumab	Complement protein C5	Paroxysmal nocturnal hemoglobinuria	
Natalizumab	α 4-integrin	Multiple sclerosis, Crohn disease	α 4-integrin: WBC adhesion Risk of PML in patients with JC virus
Ustekinumab	IL-12/IL-23	Psoriasis, psoriatic arthritis	
Other applications			
Abciximab	Platelet glycoproteins IIb/IIIa	Antiplatelet agent for prevention of ischemic complications in patients undergoing percutaneous coronary intervention	IIb times IIIa equals “ab six imab”
Denosumab	RANKL	Osteoporosis; inhibits osteoclast maturation (mimics osteoprotegerin)	Den o somab affects o steoclasts
Digoxin immune Fab	Digoxin	Antidote for digoxin toxicity	
Omalizumab	IgE	Refractory allergic asthma; prevents IgE binding to Fc ϵ RI	
Palivizumab	RSV F protein	RSV prophylaxis for high-risk infants	Pali VI zumab— VI rus
Ranibizumab, bevacizumab	VEGF	Neovascular age-related macular degeneration, proliferative diabetic retinopathy and macular edema	